Comparative Performance of Companies under NSE using Empirical Tools

Submitted in partial fulfilment of the requirements of the degree

Post-Graduate Diploma in Statistical Methods and Analytics

By

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Declaration/Undertaking

We hereby declare that the work presented in this project is carried out by us and is not copied from other sources. Wherever we have referred to the work from other sources, suitable citations and acknowledgments have been provided.

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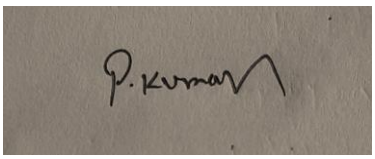

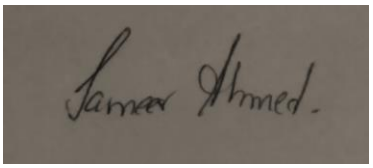

Date: 2nd Aug 2021

Certificate

This is to certify Praveen Kumar (DSTC 2021 007) and Sameer Ahmed S (DSTC 2021 010) has done the project under my supervision and guidance (from 23/04/2021 to 01/08/2021). This is an original project report based on work carried out by him / her / them in partial fulfilment of the requirement for the Post-Graduate Diploma in Statistical Methods and Analytics programme of the Indian Statistical Institute, Chennai Centre.

Prof S M Bendre

Acknowledgment

We would like to express gratitude to Prof S M Bendre for valuable guidance and suggestions during all the phases of planning, development, and completion of this project. We are thankful to her in many ways especially for generously dedicating time and supervising our project.

We also thank all of our friends who continuously helped us in shaping our thoughts in finance and stocks domain.

Finally, we thank the program and institute for giving us the opportunity to work on the project as a part of program curriculum.

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1. Introduction

National Stock Exchange (NSE) is the government owned stock exchange of India. There are many stocks across various sectors in India available for trading in NSE. For instance, sectors such as Steel, Banks, Food etc. Each sector shows its own behavioural pattern: some sectors thrive while some face a blow in a particular market window. However, it is observed that for some of the sectors such as banking, the stocks of the banks which are listed under NSE with high market capital often seem to behave in synergy i.e. the stocks of these banks often increase or decrease simultaneously. At the same time, such behaviour is not observed for some other sectors.

In this project we have selected few sectors and focus on the stock behaviour of the companies belonging to the same sector. It aims to find the dependency of companies belonging to a sector and investigate whether there is synergy among the companies. It's important to understand this synergy which will help in understanding the association of those stocks to build a robust well diverse portfolio. The thumb rule calls for construction of portfolio where the stocks are least associated. For any risk-averse investor, if there is a bad sentiment for a certain stock, its unrelated partner-stocks of portfolio will have a good chance of not getting influenced by that sentiment. This additional information on synergy and association may help in the prediction of performance on stocks and building a robust portfolio.

2. Aim

As mentioned above, we focus on the comparative behaviour of stocks of the companies belonging to the same sector, based on NSE data. In particular, the aim of the project is to see whether there is any synergy between the stocks of companies belonging to a particular sector, i.e., whether they affect each other, and if so, to quantify this association. In short, the aim is

(i) To obtain the strengths of dependency among companies belonging to a sector, for various sectors.

(ii) To understand the level of agreement of companies belonging to a particular sector.

(iii) To rank the sectors based on their association.

(iv) To also obtain the measure of association between sectors (inter-sector).

(v) To use the available information on association and dependency in the prediction of stock performances.

3. Work

3.1. Data Collection and Simplification

In a given sector, high market-cap companies are relatively less sensitive, and volatile compared to small market-cap companies. Hence companies with high market capitalisation (market-cap) belonging to a sector in NSE are selected. We selected 8 sectors and four highest market-cap companies from each sector. For each company, prices both at the opening time and closing time for each day are collected for past 10 years (from 1st April 2011 to 31st March 2021). Our data source is Yahoo Finance News. Since the stock market is closed on weekends and some public holidays, the figures were not available for all the days. We considered the data deleting these closed days. This led to getting stock values for 2477 days for each company, giving a time series data of length 2477.

Our interest is only in the increasing or decreasing stock price of the company on consecutive days and not the amount of change in the stock price. Co-jumps are correlated jumps which talk about the simultaneous direction of stocks. Hence for each company the stock prices were converted to the binary form {0, 1} where

0 := decrease in the stock price compared to the previous day

1 := increase in the stock price compared to the previous day

In our work, we have only considered that the dependence of company's performance today will only depend on the performance on its previous day (Discrete-Time Markov Chain).

There is a concept of 'co-jumps' which is getting attention in the literature lately (M Arouri, 2019) where the actual stock prices of two or more companies are studied. Our aim here is only in the 'direction' of stock prices of different companies.

By converting the data to binary form, we get a discrete time discrete space Markov Chain with state space {0, 1} and length 2477 for each company. To avoid the sparsity of the data we have assumed that a company's performance on any day will only depend on its performance on the previous day and have focused on the first-order Markov Chain in this work.

3.2. Sector-wise Performance of Markov Chains

To illustrate our investigations, we consider only Banking sector at present. For the Banking Sector, the top four companies (banks) based on the highest market-cap are AXIS, ICICI, Kotak-Mahindra and HDFC Banks.

To begin with we considered HDFC bank and computed the empirical transition probability matrix (TPM). The empirical TPM is

HDFC	**X_1=1**	**X_1=0**
X_0=1	**0.477 (572)**	**0.523 (627)**
X_0=0	**0.493 (627)**	**0.507 (650)**

where X_0 stands for the state of the Markov chain on the previous day and X_1 stands for state of the Markov chain on the current day. Thus the above TPM gives us the empirical estimates of $P[X_1 = 1| X_0 = 1] = 0.47$ and so on. The figures in the brackets give the actual number of transitions, there are 572 number of transitions occurring

from state $X_0=1$ to state $X_1=1$. As we can see, all the probabilities are around 0.5 and not giving any significant information on the behaviour of the stock prices.

Analogous conversions and computations for the other three companies, namely AXIS Bank, ICICI Bank and Kotak Mahindra Bank yielded the empirical TPMs which are presented below. As can be seen, all the transition probabilities are close to 0.5, as anticipated.

AXIS	$X_1=1$	$X_1=0$	KOTAK	$X_1=1$	$X_1=0$	ICICI	$X_1=1$	$X_1=0$
$X_0=1$	0.463	0.537	$X_0=1$	0.443	0.557	$X_0=1$	0.459	0.541
$X_0=0$	0.457	0.543	$X_0=0$	0.516	0.484	$X_0=0$	0.467	0.533

We further aim to combine the available information. Suppose we have N companies from each sector, each giving a 2477 length Markov chain with state space consisting of two values {0, 1}. Since the aim is to investigate the dependence structure among these Markov chains, we need to consider a N×1 multivariate Markov chain with 2^N states. The larger value of N may lead to better information about the dependence between the stock prices of the companies. However, the Markov chains are of length 2477 each and hence we restricted our study to N=4, as mentioned in Section 3.1.

The empirical TPM of this 4×1 multivariate Markov chain with 2^4 states corresponding to the four banks AXIS, ICICI, Kotak-Mahindra and HDFC is computed based on the data of these four banks and is observed to be

TPM	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Y_8	Y_9	Y_{10}	Y_{11}	Y_{12}	Y_{13}	Y_{14}	Y_{15}	Y_{16}
Y_1	0.16	0.05	0.06	0.05	0.03	0.06	0.04	0.02	0.02	0.03	0.05	0.06	0.05	0.04	0.07	0.21
Y_2	0.16	0.01	0.04	0.02	0.08	0.1	0.03	0.02	0.01	0.02	0.03	0.06	0.07	0.08	0.05	0.24
Y_3	0.16	0.06	0.03	0.06	0.04	0.09	0.02	0.02	0.02	0.05	0.04	0.06	0.07	0.08	0.02	0.18
Y_4	0.19	0.05	0.06	0.02	0.03	0.04	0.03	0.04	0.02	0.04	0.04	0.06	0.08	0.06	0.09	0.17
Y_5	0.13	0.06	0.03	0.03	0.06	0.07	0.01	0.02	0.01	0.03	0.07	0.1	0.07	0.06	0.07	0.21
Y_6	0.17	0.05	0.04	0.03	0.03	0.06	0.02	0.03	0.03	0.03	0.09	0.04	0.04	0.1	0.06	0.17
Y_7	0.15	0	0.06	0.04	0.01	0.04	0.03	0.06	0.03	0.01	0.06	0.07	0.04	0.1	0.1	0.21
Y_8	0.12	0.04	0.01	0.06	0.01	0.09	0.01	0.04	0	0.07	0.07	0.01	0.01	0.12	0.1	0.19
Y_9	0.17	0.03	0.03	0.03	0	0.12	0.02	0.03	0.02	0.05	0.09	0.05	0.06	0.03	0.11	0.17
Y_{10}	0.17	0.07	0.07	0.04	0.01	0.07	0.04	0.03	0.03	0.03	0.09	0.06	0.01	0.06	0.04	0.17
Y_{11}	0.15	0.06	0.01	0.05	0.04	0.06	0.03	0.02	0.04	0.02	0.08	0.04	0.09	0.05	0.06	0.22
Y_{12}	0.15	0.02	0.03	0.05	0.07	0.03	0.03	0.02	0.03	0.02	0.08	0.07	0.06	0.07	0.04	0.25
Y_{13}	0.17	0.03	0.05	0.04	0.05	0.01	0.07	0.05	0.03	0.01	0.09	0.04	0.03	0.08	0.06	0.2
Y_{14}	0.15	0.05	0.04	0.03	0.05	0.06	0.03	0.02	0.03	0.04	0.08	0.03	0.04	0.03	0.09	0.23
Y_{15}	0.17	0.03	0.07	0.09	0.03	0.03	0.03	0.04	0.03	0.01	0.11	0.04	0.06	0.03	0.06	0.15
Y_{16}	0.2	0.04	0.05	0.04	0.06	0.06	0.02	0.02	0.03	0.03	0.07	0.03	0.05	0.07	0.05	0.18

where Y_i, i=1, 2, …, 16 is a 4×1 vector where each element in vector represents the direction of the company stock in terms of 0 and 1. In particular, we define

	AXIS	ICICI	KOTAK	HDFC
Y_1	1	1	1	1
Y_2	1	1	1	0
Y_3	1	1	0	1
Y_4	1	0	1	1
Y_5	0	1	1	1
Y_6	1	1	0	0
Y_7	1	0	1	0
Y_8	1	0	0	1
Y_9	0	1	1	0
Y_{10}	0	1	0	1
Y_{11}	0	0	1	1
Y_{12}	1	0	0	0
Y_{13}	0	1	0	0
Y_{14}	0	0	1	0
Y_{15}	0	0	0	1
Y_{16}	0	0	0	0

Thus, for instance, cell (3,4) of TPM gives the proportion of the times $Y_4 = (1, 0, 1, 1)$ was observed on the next day, given that the today's observation was $Y_3 = (1, 1, 0, 1)$. From the 16×16 empirical TPM, one can clearly see high probabilities in first and last column. First and last columns are the states where company stocks move all up and all down (Synergy). It suggests that for the banking sector, no matter what state (Y_i) the sector is in on the previous day, it has higher probability to go to a state where all the stocks go up (Y_1) or all go down (Y_{16}) on the current day. This indicates some form of association or dependence between the stock prices of four highest market-cap banks.

We performed analogous computations on the data from 7 other sectors, The table below, displays the selected four companies based on market capitalisation of corresponding sectors:

Sl. No	**Sectors**	**Top 4 companies based on high market-cap**			
1	**Automotive**	Maruti Suzuki	Bajaj Auto	Tata Motors	M & M
2	**Food**	Nestle	Britannia	Tata Cons.	Uni. Spirits
3	**Iron**	JSW Steel	Tata Steel	NMDC	Hindalco
4	**IT**	TCS	Infosys	Wipro	Hcl Tech
5	**Media**	Sun Tv	Zee Enter	PVR	Tv18
6	**Pharma**	Sun Pharma	Divis Lab	Dr Reddys	Cipla
7	**Fuels**	BPCL	IOCL	ONGC	Reliance

For each of the four companies in each sector, the individual empirical 2×2 TPM showed all four values close to 0.5, thus indicating that it is difficult to speculate about the stock price on the next day, given the stock price on the previous day, as expected. However, the pattern of the 16×16 TPM for each sector was different. Few sectors such as Iron and Steel, IT etc. showed the pattern of having higher probabilities for going to synergy states (Y_1 and Y_{16}), while other sectors such as Food and Media sectors did not.

3.3. Ranking Sectors based on Synergy using TOPSIS Method

To find a measure of synergy for a sector, we defined three criteria based on the direction of the stocks namely:

(A) Full Synergy (All companies in a sector having same direction of stocks, either all increasing or all decreasing, i.e. $\{Y_{1,}\ Y_{16}\}$)

(B) Partial Synergy (3 out of 4 companies having one direction and the other company having the opposite direction, i.e. $\{Y_2, Y_3, Y_4, Y_5, Y_{12}, Y_{13}, Y_{14}, Y_{15}\}$)

(C) No Synergy (2 companies having one direction and other two the opposite direction. $\{Y_6, Y_7, Y_8, Y_9, Y_{10}, Y_{11}\}$)

Criteria	Automotives	Banks	Food	Iron	IT	Media	Pharma	Fuels
Full Synergy	722	903	553	982	853	599	715	808
Partial Synergy	1150	983	1168	1020	1098	1185	1114	1081
No Synergy	605	591	756	475	526	693	648	588

The table shows the number of days out of 2477 which fall in a defined synergy criterion for the 8 selected sectors. Our interest is to rank these sectors based on their Synergy.

We use TOPSIS method (Rahim, 2018) which is a multi-criteria decision-making method mainly used to rank the subjects; where by subject, we mean a sector here. We will first categorise the criteria as benefit features (Higher the value, better the synergy) and cost features (Lower the value, better the synergy) accordingly. The TOPSIS method ranks subjects using the distances of a subject from positive ideal solution and also from negative ideal solution.

Positive ideal solution is the vector of values consisting the highest number of days for benefit features and lowest number of days for cost features. Negative ideal solution is the vector of values consisting the lowest number of days for benefit features and highest number of days for cost features. Best case for a subject is to have shortest distance from positive ideal solution and farthest from negative ideal solution.

Along with 8 selected sectors, we also added an independent subject, where all criteria are equally likely. Thus, we shall have one more column of independent subject, which will be treated as reference or baseline.

First criteria (A) have two states $\binom{4}{0} + \binom{4}{4} = 2$

Second criteria (B) have 8 states $\binom{4}{1} + \binom{4}{3} = 8$

Third criteria (C) have 6 states $\binom{4}{2} = 6$.

Hence, we have split 2477 days for baseline subject into the proportions mentioned above.

Criteria	Automotive	Banks	Food	Iron	IT	Media	Pharma	Fuels	Baseline
Full Synergy	722	903	553	982	853	599	715	808	309.625
Partial Syner	1150	983	1168	1020	1098	1185	1114	1081	1238.5
No Synergy	605	591	756	475	526	693	648	588	928.875

Procedure:

i. Create a table with subjects as rows and criteria as columns. Thus, we have nine rows and three columns here. We represent rows by j and columns by i.

Assign weightages to each criterion, we assigned weights w_1= 0.5 to Full synergy; w_3= 0.5 to No synergy and w_2= 0 to Partial synergy. As the full synergy criterion is a benefit feature, No synergy criterion is a cost feature and Partial synergy criterion is neither benefit feature nor cost feature.

	Full Association (0 or 4)	Partial Association (1 or 3)	No Association (2)
Automotives	722	1150	605
Banks	903	983	591
Food	553	1168	756
Iron and Steel	982	1020	475
IT	853	1098	526
Media	599	1185	693
Pharma	715	1114	648
Fuels	808	1081	588
Baseline	309.625	1238.5	928.875

ii. Transform the frequencies presented in the table above to their normalised form.

$$r_{ij} = \frac{f_{ij}}{\sqrt{\sum_{j=0}^{J} f_{ij}^2}}$$

where j represents rows j = 1, …, J and i represents columns, ; i= 1, …, n . We have J=9 and n = 3 here.

iii. Multiply weights to the newly transformed values.

$v_{ij} = w_i r_{ij}$,

where j = 1, …, J, i = 1, …, n and w_i are the weights assigned to corresponding features. We have $w_1 = w_3 = 0.5$ and $w_2 = 0$

iv. Determine the ideal and negative ideal solution by defining

$$A^* = \{v_1^*, \ldots, v_n^*\}$$
$$= \{(\max_j v_{ij} \mid i \in \mathrm{I}'), (\min_j v_{ij} \mid i \in \mathrm{I}'')\}$$
$$A^- = \{v_1^-, \ldots, v_n^-\}$$
$$= \{(\min_j v_{ij} \mid i \in \mathrm{I}'), (\max_j v_{ij} \mid i \in \mathrm{I}'')\}$$

where I′ is associated with benefit feature and I″ is associated with cost feature.

v. Calculate distance of every subject from ideal solution and store it as D^*

$$D_j^* = \sqrt{\sum_{i=1}^{n} (v_{ij} - v_i^*)^2} \quad , j=1, \ldots, J$$

vi. Similarly, calculate the distance of subjects from negative-ideal solution as D^-

$$D_j^- = \sqrt{\sum_{i=1}^{n} (v_{ij} - v_i^-)^2} \quad , j=1, \ldots, J$$

Calculate relative closeness (C^*) of distances of each subject.

$$C_j^* = \frac{D_j^-}{(D_j^* + D_j^-)} \quad , j=1, \ldots, J$$

vii. Rank the subjects using C^*, where higher C^* implies better synergy or association.

After applying the above procedure, we found the ranks of the sectors as follows.

Subjects	C^*	Rank
Automotive	0.649	5
Banks	0.822	3
Food	0.369	8
Iron and Steel	1.000	1
IT	0.834	2
Media	0.463	7
Pharma	0.609	6
Fuels	0.745	4
Baseline	0.000	9

It can be seen from the table above that as defined, the index C^* of baseline is 0.

The index C^* =1 for the Steel sector (Iron and Steel) is highest among all the selected sectors and we can conclude that the four companies from Iron and Steel sector with high market-cap have the highest levels of association (in terms of their stock movements among the selected sectors). Selected four companies from banking sector have the second highest levels of association. Companies from Food sector have least levels of association while the companies from Media sector have second least levels of association.

3.4. Dependency of Sectors using Markov Chain.

Continuing from Section 3.2., where we used empirical TPM of a multivariate Markov chain to understand synergy level of a sector, we now investigate the dependency of companies in a sector using the TPMs.

To quantify the dependency, we use Kulback-Leibler (KL) divergence rate as a criterion to compute the distance between the empirical TPM and the TPM under the assumption of independence of the 4 Markov chains for each sector.

Let TPM A denote the empirical TPM of a sector based on the actual data. We then calculate TPM of a sector assuming independence of all companies in that sector and call it TPM B. The KL divergence rate between these two matrices gives us an idea of how far the TPM A lies with respect to TPM B.

TPM A is already calculated for each sector as methodology used in Section 3.2.

To calculate the TPM B for a sector we carry out the following steps:

i. Assume the independence of 4 companies in a sector.

ii. Calculate empirical TPM (1 dimensional) for each company (univariate MC)

iii. Calculate the probabilities of (4 dimensional) TPM B by multiplying corresponding values of TPM of each company. For example, if we are to calculate probability to go from state $Y_1 = (1,1,1,1)$ to state $Y_{10} = (0,1,0,1)$ then we do it by multiplying the empirical probabilities

$P[X_{11}=0|X_{10}=1]\times P[X_{21}=1|X_{20}=1]\times P[X_{31}=0|X_{30}=1]\times P[X_{41}=1|X_{40}=1]$

where X_{i1} represents the stock status of ith company on the current day and X_{i0} is the stock status of the same company on the previous day, i=1, …,4.

iv. Thus we obtain a 16×16 empirical TPM B

Now that we have TPM B which is TPM of sector assuming independence, we need to calculate the KL divergence rate for each TPM A and TPM B. Rached et al (2004) have proposed the Kullback-Leibler divergence rate between two first order time-homogeneous Markov chains with unique stationary distributions. We extend this by using the following formula. Let

$$S(U_2|U_1 = k) \triangleq \sum_{j \in ¥} p_{kj}\log\left(\frac{p_{kj}}{q_{kj}}\right)$$

where $U_1, U_2 \in ¥ = \{Y_1, Y_2 \ldots\ldots Y_{16}\}$ are all possible states of the multivariate Markov Chain, p_{kj} and q_{kj} are probabilities corresponding to the kth row and j^{th} column for TPM A and TPM B respectively. We assume that $p_{kj} = 0$ if $q_{kj} = 0$, for all k, j $\in$ ¥. Further the approximate KL divergence is

$$D(A\|B) = \sum_{k \in ¥} \pi_k\, S(U_2|U_1 = k)$$

where $\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots\ldots \pi_{16})$ are the unique stationary distribution probabilities of TPM A. Note that the stationary distribution for TPM A is a multivariate (4 dimensional) probability mass function for a sector. For example, π_2 is obtained by computing the probability of the multivariate Markov chain belonging to state Y_2 in the long run (asymptotically).

After following the procedure mentioned above, we calculated the KL divergence rate for all the sectors and ranked them in descending order.

Rank	Sectors	KL Divergence Rate
1	Iron and Steel	0.312
2	Banks	0.256
3	IT Service	0.231
4	Fuels	0.217
5	Pharma	0.155
6	Automotive	0.148
7	Media	0.101
8	Food	0.085

From the above table, Iron and steel sector has highest KL Divergence rate among all the sectors. It implies that, in the given group of 8 sectors, Companies in Steel sector and Banks sector have highest and second highest levels of dependency based on KL Divergence rates. Companies in Food sector and Media sector have least and second least levels of dependency based on KL Divergence rates.

3.5. Test for Dependency

Under the assumption of independence of the four Markov chains, the KL divergence rate D, say, is =0. Since D is a non-negative random variable, it will be significantly larger than 0 in case of dependent Markov chains. Hence to test

H_0 : The four Markov chains are independent

against

H_1 : There is dependence between the four Markov chains,

is equivalent to testing

H_0: $D = 0$ against $H_1 : D > 0$.

Since the distribution of D is unknown, we decided to compute the empirical distribution of D and then calculate empirical p-value for each sector to test for independence.

For each sector, in order to obtain empirical distribution, we calculate KL Divergence rate (d_i) among a new randomly generated TPM (TPM C) and TPM B.

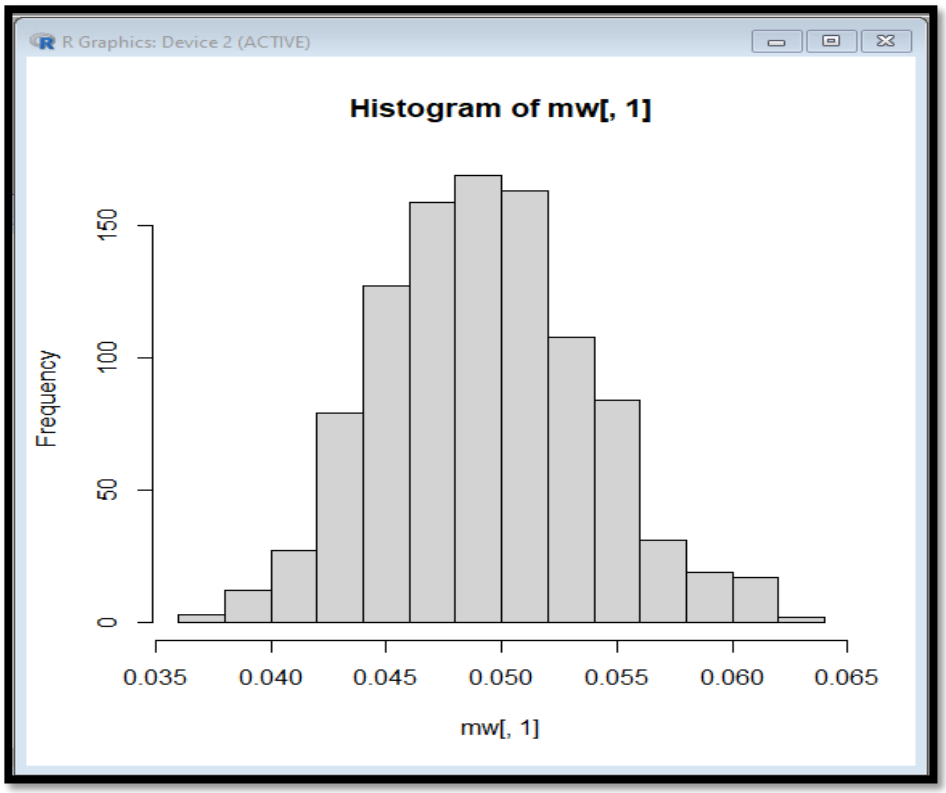
To construct TPM C for a sector:

i. First, we need to generate Markov Chain Sequence of length n=2477 matching our original data from 1D TPM of each company.

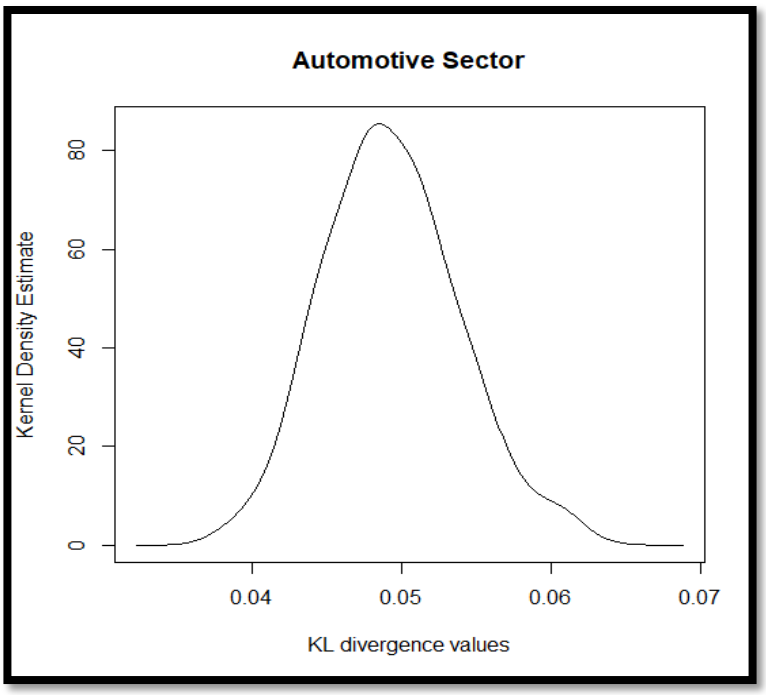
ii. Since there is independence, we concatenate sequences of all 4 companies to obtain a Markov Chain Sequence (4D) where each element (Y) is 4 X 1 vector [X1, X2, X3, X4] on a given day. X_i is the direction of ith company for a given day generated from the above point.

iii. We find 4D TPM (TPM C) using the Markov Chain Sequence (4D) generated from step ii.

We calculate 'M' numbers of TPM C by performing the above steps 'i' times, we calculate d_m for M^{th} simulation.



The above picture is the histogram for d_m for Automotive sector.



We have also plotted Kernel Density estimate for KL Divergence rate values of Automotive sector, giving us a rough idea on empirical distribution. Our objective is to test for dependency. We calculated empirical p-values using P [d_m > calculated value]. Our calculated value is 0.148 for Automotive sector. We found empirical p-value as 0 since no d_i is greater than 0.148 in M=1000 simulations.

Hence empirical p-value is 0 and hence we reject null hypothesis. Companies among Automotive sector is dependent. We performed all the above steps for all 8 sectors and we found that all sectors have empirical p-value = 0. Hence all sectors are dependent.

3.6. Association between Sectors.

In addition to investigating the inter-sector dependence structure, we also look at the dependence between different sectors. For this intra-sector dependence, we considered 2 sectors to compute measure of association between them (pair-wise association). For every sector with four companies, we have defined 5 features namely, the stocks of

(A) All companies going down.

(B) One company going up and remaining three going down.

(C) Two companies going up and remaining two going down.

(D) Three companies going up and remaining one going down.

(E) All companies going up.

To measure association between two sectors, we created a contingency table of 5×5 cells. Row represents one Sector, column represents another Sector and all features (A to E) are ordinal variables. The cell values represent the number of days.

For instance, contingency table for Automotive and Banking sector is as follows

		Banking Sector				
		A	B	C	D	E
Automotive Sector	A	178	111	84	31	20
	B	162	171	168	85	60
	C	100	151	156	125	73
	D	36	95	129	115	128
	E	14	27	54	72	131

where, for instance, the number 178 for cell (1,1) indicates that for all four companies of both these sectors, the observation was (0, 0, 0, 0) on 178 days.

We observe that the diagonal elements have higher values, which gives us some idea about association. Since the features are ordinal in nature, we use Goodman- Kruskal Gamma measure to find association between two sectors.

Steps involved in calculation of Gamma:

i. We have to find number of concordant pairs (Nc) with respect to each cell. Consider cell T_{lm} (l^{th} row and m^{th} column), its concordant pair is with all cell values below the column of cell and right to the row of cell, i.e. cells T_{ij} where i>l and j>m. Then

   Nc $=\sum_{l=1}^{l=5}\sum_{m=1}^{m=5} T_{lm} \times (\sum_{i>l}^{5}\sum_{j>m}^{5} T_{ij})$

ii. We have to also find number of discordant pairs (Nd) with respect to each cell. Consider cell T_{lm} (l^{th} row and m^{th} column), its discordant pair is with all cell values below the column of cell and left to the row of cell, T_{ij} where i<l and j>m.

   Nd $=\sum_{l=1}^{l=5}\sum_{m=1}^{m=5} T_{lm} \times (\sum_{i<l}^{5}\sum_{j<m}^{5} T_{ij})$

iii. Gamma coefficient is

$$\gamma = \frac{(\mathrm{Nc}-\mathrm{N_d})}{(\mathrm{Nc}+\mathrm{N_d})}$$

By definition , the Gamma coefficient varies from -1 to +1where the magnitude indicates the strength and sign indicates the nature of association.

For Automotive and Banking sector Gamma coefficient is 0.429. The Gamma coefficients for different pairs of sectors is presented in the table below.

	Automotive	**Banks**	**Food**	**Iron and Steel**	**IT**	**Media**	**Pharma**	**Fuels**
Automotive	_	0.43	0.321	0.423	0.189	0.374	0.284	0.336
Banks	0.43	_	0.304	0.407	0.167	0.339	0.262	0.377
Food	0.321	0.304	_	0.298	0.163	0.326	0.303	0.25
Iron and Steel	0.423	0.407	0.298	_	0.137	0.366	0.244	0.382
IT	0.189	0.167	0.163	0.137	_	0.149	0.205	0.163
Media	0.374	0.339	0.326	0.366	0.149	_	0.239	0.299
Pharma	0.284	0.262	0.303	0.244	0.205	0.239	_	0.251
Fuels	0.336	0.377	0.25	0.382	0.163	0.299	0.251	__

Since Gamma coefficient is symmetric in nature, we obtained the symmetric matrix of associations. Highest Gamma coefficient is found between Automotive and banking sector implying highest inter-sector dependence amongst these two sectors, least gamma coefficient is obtained between Steel and IT sector implying weak inter-sector dependence of all pairwise comparisons computed.

3.7. Prediction of a company performance using Hidden Markov Model

Next, our objective is to predict the company's direction of the company stock but not the amount of change, given some additional information.

Along with the stock based binary index of each company, we used Sensex index as the additional information. We are interested to see if we are improving the chance of predicting the direction (increasing or decreasing) of company's stock tomorrow given the direction of Sensex index (increasing or decreasing) today.

We used the Sensex direction today as our observed state, which can give us additional knowledge on Company's stock direction tomorrow (hidden state).

Combination (I): Sensex direction today as an observable state and direction of Axis bank stock tomorrow as a hidden state.

Sensex\Axis Bank	Y_1=1	Y_1=0
X_1=1	0.455	0.545
X_1=0	0.435	0.565

The above shown is empirical emission transition probability matrix where X_1 is a random variable representing the sensex direction today and Y_1 is a random variable representing the Axis bank stock direction tomorrow. As we can see this combination of using Sensex today as hidden state for predicting he stock of the Axis bank tomorrow is not that effective as probabilities are close to 0.5, similar to the empirical TPM of 1dimensional Markov chain itself.

Combination (II): Sensex direction past 3 days (today, yesterday and day before yesterday) as observable states and Axis bank tomorrow as hidden state.

Sensex\Axis Bank	Y_1=1	Y_1=0
Z_1 (1,1,1)	0.493	0.507
Z_2 (0,1,1)	0.460	0.540
Z_3 (1,0,1)	0.451	0.549
Z_4 (0,0,1)	0.419	0.581
Z_5 (1,1,0)	0.470	0.530
Z_6 (0,1,0)	0.443	0.557
Z_7 (1,0,0)	0.474	0.526
Z_8 (0,0,0)	0.476	0.524

The above shown is empirical emission transition probability matrix where Z_1 is a 3×1 random vector representing today, yesterday and day before yesterday's sensex direction respectively and Y_1 is a random variable representing the Axis bank stock direction tomorrow. As we can see this combination (II) is not that effective as probabilities are close to 0.5.

Combination (III): Sensex direction past 3 days (today, yesterday and day before yesterday) as observable state and Axis bank direction past 3 days (today, yesterday and day before yesterday) is hidden state.

Sensex\Axis Bank	Z_1 (1,1,1)	Z_2 (0,1,1)	Z_3 (1,0,1)	Z_4 (0,0,1)	Z_5 (1,1,0)	Z_6 (0,1,0)	Z_7 (1,0,0)	Z_8 (0,0,0)
Z_1 (1,1,1)	0.413	0.429	0.323	0.500	0.471	0.467	0.474	0.333
Z_2 (0,1,1)	0.568	0.338	0.375	0.350	0.400	0.366	0.545	0.429
Z_3 (1,0,1)	0.484	0.333	0.440	0.489	0.500	0.625	0.439	0.565
Z_4 (0,0,1)	0.500	0.436	0.529	0.475	0.313	0.381	0.455	0.500
Z_5 (1,1,0)	0.478	0.500	0.800	0.636	0.500	0.580	0.422	0.625
Z_6 (0,1,0)	0.435	0.531	0.625	0.536	0.482	0.408	0.652	0.510
Z_7 (1,0,0)	0.381	0.733	0.431	0.536	0.317	0.533	0.372	0.422
Z_8 (0,0,0)	0.636	0.393	0.526	0.554	0.455	0.373	0.452	0.487

Cell values of the above table is the probability of Axis bank stock direction going up tomorrow, given the states of Sensex direction past 3 days (Z_i) and Axis bank direction past 3 days (Z_i). We can see that few probability values are away from 0.5 and we hope to use this mbination for improving the prediction of Axis bank stock direction tomorrow using Hidden Markov Model. (HMM).

We are using Axis bank (1D TPM) as empirical TPM but with lag 3. We are also taking emission probability matrix between Sensex and Axis bank with lag 3.

Axis\Axis Bank	Y_t=1	Y_t=0
Y_{t-3}=1	0.458	0.542
Y_{t-3}=0	0.470	0.530

Empirical TPM (lag 3) Row represents (Y_{t-3}) the direction of Axis bank day before yesterday and column represent (Y_t) direction of Axis bank tomorrow.

Sensex\Axis Bank	Y_t=1	Y_t=0
X_{t-3}=1	0.450	0.550
X_{t-3}=0	0.475	0.525

Empirical emission probability matrix (lag 3) Row represents (X_{t-3}) the direction of Sensex day before yesterday and column represent (Y_t) direction of Axis bank tomorrow.

We used Viterbi algorithm to predict the Y_t given these two probability matrices. It did not give satisfactory results because, one can see that no matter what observed state is i.e., going up or down, the probability is higher for $Y_t = 0$. Hence all predictions end up giving $Y_t = 0$.

4. Results and Conclusions.

1) Steel sector companies have highest level of agreement (Synergy) whereas food sector companies have the least Synergy among the selected sectors.
2) Steel sector companies have highest dependency amongst themselves where as food sector companies have least dependency amongst themselves in the selected sectors and no sector is found independent.
3) Highest association is found between Automotive and Banking Sector, least association is found between IT and Steel Sector.
4) Sensex Index wasn't useful enough to predict company's direction using HMM.

5. References:

- Alajaji, F., Rached, Z., 2004. The Kullback–Leibler Divergence Rate Between Markov Sources, IEEE Transactions on Information Theory.
- Andy, P.U., Andysah, P.U.H., Khairunissa, K., Mufida, K., Rahim, R., Siti A., Siti, S., Supiyandi, S., Tri, L., Wiwit, A.T., Yudie, I., 2018. TOPSIS Method Application for Decision Support System in Internal Control for Selecting Best Employees, Journal of Physics Conference Series.
- Arouri, M., Duc K.N., Kuntara, P., Oussama, M., Cojumps and Asset Allocation in International Equity Markets, Journal of Economics Dynamics and Control, Forthcoming.
  - Source of stocks data: https://in.finance.yahoo.com

Appendix: R codes

Packages installed in R code.

```
install.packages(‘DescTools’)
install.packages(‘vcd’)
install.packages(‘vcdExtra’)
install.packages(‘expm’)
install.packages(‘HMM’)
install.packages(‘markovchain’)
install.packages(‘KernSmooth’)

library(DescTools)
library(vcd)
library(vcdExtra)
library(expm)
library(HMM)
library(markovchain)
library(KernSmooth)
```

A1. Code for Transition Probability Matrix B

```
indmat=function(f1){
m1=matrix(c(1,1,1,1,1,1,1,0,1,1,0,1,1,1,0,0,1,0,1,1,1,0,1,0,1,0,0,1,1,0,0,0,0,1,1,1,0,1,1
,0,0,1,0,1,0,1,0,0,0,0,1,1,0,0,1,0,0,0,0,1,0,0,0,0),ncol=4,byrow=TRUE)
m1
ind=matrix(rep(0,256),ncol=16)
t=pr(f1)
for (i in 1:16) {
for (j in 1:16) {
z2=c(1,1,1,1)
for (k in 1:4) {
if(m1[i,k]==1&m1[j,k]==1) {
z2[k]=t[k,1]
}
else if(m1[i,k]==1&m1[j,k]==0) {
z2[k]=t[k,2]
}
else if(m1[i,k]==0&m1[j,k]==1) {
z2[k]=t[k,3]
}
else {
z2[k]=t[k,4]
} }
ind[i,j]=prod(z2)
```

```
} }
return(ind)
}
```

A2. Code for Transition Probability matrix (1 Dimensional)

```
#11,10,01,00
pr=function(f2) {
f2=as.data.frame(f2)
n=(length(f2[,1]))-1
zi=matrix(rep(0,16),ncol=4)
for (i in 1:4) {
c1=0; p1=0; c2=0; p2=0
for (j in 1:n) {
if (is.na(f2[j,i])|is.na(f2[j+1,i])){
r=1
}
else{
if (f2[j,i]==1&f2[j+1,i]==1) {
c1=c1+1
}
if(f2[j+1,i]==1) {
p1=p1+1
}
if(f2[j,i]==0&f2[j+1,i]==1) {
c2=c2+1
}
if(f2[j+1,i]==0) {
p2=p2+1
} } }
zi[i,1]=c1/p1; zi[i,2]=1-(c1/p1); zi[i,3]=c2/p2; zi[i,4]=1-(c2/p2)
}
return (zi)
}

#CODE FOR 1D TPM previous day for j,current day for k
conmat=function(ba,j=1,k=2) {
prop=matrix(c(0,0,0,0),nrow=2)
for (i in 1:(length(ba[,j])-1)) {
if(ba[i+1,j]==0&ba[i,k]==0) {
prop[2,2]=prop[2,2]+1
}
else if(ba[i+1,j]==1&ba[i,k]==1) {
prop[1,1]=prop[1,1]+1
}
else if(ba[i+1,j]==1&ba[i,k]==0) {
```

```
prop[2,1]=prop[2,1]+1
}
else {
prop[1,2]=prop[1,2]+1
} }
return (t(prop))
}

#1d TPM 11,10,01,00
pr1=function(f2) {
n=(length(f2)-1)
zi=matrix(rep(0,4),ncol=4)
c1=0; p1=0; c2=0; p2=0
for (j in 1:n) {
if (is.na(f2[j])|is.na(f2[j+1])){
r=1
}
else{
if (f2[j]==1&f2[j+1]==1) {
c1=c1+1
}
if(f2[j+1]==1) {
p1=p1+1
}
if(f2[j]==0&f2[j+1]==1) {
c2=c2+1
}
if(f2[j+1]==0) {
p2=p2+1
} }
zi[1]=c1/p1; zi[2]=1-(c1/p1); zi[3]=c2/p2; zi[4]=1-(c2/p2)
}
return (zi)
}
```

A3. Code for Transition Probability matrix (4 Dimensional)

```
cnv=function(f3){
m1=matrix(c(1,1,1,1,1,1,1,0,1,1,0,1,1,1,0,0,1,0,1,1,1,0,1,0,1,0,0,1,1,0,0,0,0,1,1,1,0,1,1
,0,0,1,0,1,0,1,0,0,0,0,1,1,0,0,1,0,0,0,0,1,0,0,0,0),ncol=4,byrow=TRUE)
a=rep("Y",16)
b=1:16
a=paste(a,b,sep="")
n=(length(f3[,1]))
cn=rep(0,n)
```

```
for (i in 1:n) {
for (j in 1:16) {
if(isTRUE(all.equal(m1[j,],as.numeric(f3[i,])))) {
cn[i]=a[j]
} } }
return (TPM(cn))
}

TPM=function(f4) {
a=rep("Y",16)
b=1:16
a=paste(a,b,sep="")
z=1
TP=matrix(rep(0,256),nrow=16)
while(z < length(f4)) {
for (i in b) {
for (j in b) {
if ((f4[z+1]==a[i])&((f4[z])==a[j])) {
TP[i,j]=TP[i,j]+1
} } }
z=z+1
}
KP=matrix(rep(0,256),nrow=16)
for (i in b) {
for (j in b) {
KP[i,j]=TP[i,j]/sum(TP[i,])
} }
return(KP)
}
```

A4. Code for Kulback-Leibler Divergence rate

```
mkvd=function(a,b){
a1=a%^%10:
s2=0
for (i in 1:16) {
s1=0
for (j in 1:16) {
if((a[i,j]!=0)&(b[i,j]!=0)){
s1=a[i,j]*log(a[i,j]/b[i,j])+s1
} }
s2=a1[1,i]*s1+s2
}
return(s2)
}
```

```
d=read.csv("C:/Users/TEST USER/Desktop/Project/Compiled data/All sorted
Data.csv") #Data

k=c(1,2,3,4)
dis=rep(1,8)
for (i in 1:8) {
dis[i]=mkvd(cnv(d[,k]),indmat(d[,k]))
k=k+4
}
Dis #KL Distances
```

A5. Code for simulation of Markov Chain

```
indsim2=function(da){
si=matrix(rep(0,2477*4),ncol=4)
for (i in 1:4) {
statesNames=c(1,0)
MCA=new("markovchain",transitionMatrix=matrix(pr1(da[,i]),nrow=2,byrow=TRUE
, dimnames=list(statesNames,statesNames)))
h=markovchainSequence(n=2477,markovchain = MCA, t0 = sample(0:1,1))
for (k in 1:2477) {
si[k,i]=h[k]
} }
return(cnv(si))
}
#indsim2(d[,9:12])

mw=matrix(rep(0,8000),ncol=8); o=1:4
for (k in 1:8) {
dw=cnv(d[,o])
e2=indmat(d[,o])
for (i in 1:1000) {
mw[i,k]=mkvd(indsim2(d[,o]),e2)
}
o=o+4
}
write.csv(as.data.frame(mw),"KL.csv") #KL distances d_i

#KDE plot
mw1=read.csv("C:/Users/TEST USER/Desktop/Project/Compiled data/KLval.csv")
head(mw1)
est=bkde(mw1[,2], kernel="normal")
plot(est,type='l',ylab='Kernel Density Estimate',xlab='KL divergence values',
main="Automotive Sector")
```

A6. Code for Hidden Markov Model

Combination I

```
# se=read.csv("C:/Users/TEST USER/Desktop/Project/Compiled data/Eprob Se
R.csv")
#Sensex past day with a particular company
eprob1=function(da=d,w=1,q=7) {
a1=expand.grid(1:0); TP=matrix(rep(0,4),nrow=2)
for (i in 1:2477){
for (j in 1:2){
if(isTRUE(all.equal(da[i,q],as.numeric(a1[j,])))) {
if(isTRUE(all.equal(da[i+1,w],1))) { TP[j,1]=TP[j,1]+1 }
else { TP[j,2]=TP[j,2]+1 } } } }
KP=matrix(rep(0,4),nrow=2)
for (i in 1:2) {
for (j in 1:2) {
KP[i,j]=TP[i,j]/sum(TP[i,])
} }
return(KP) }
eprob1(se,1,6)
```

Combination II

```
#Sensex past 3 days with a particular company
eprob=function(da=d,w=1,q=7) {
a1=expand.grid(1:0,1:0,1:0)
TP=matrix(rep(0,16),ncol=2)
for (i in 2:2474){
for (j in 1:8){
if(isTRUE(all.equal(da[i:(i+2),w],as.numeric(a1[j,])))) {
if(isTRUE(all.equal(da[i-1,q],1))) { TP[j,1]=TP[j,1]+1 }
else { TP[j,2]=TP[j,2]+1 } } } }
KP=matrix(rep(0,16),nrow=8)
for (i in 1:8) {
for (j in 1:2) {
KP[i,j]=TP[i,j]/sum(TP[i,])
} }
return(KP) }
l1=4
eprob(se,1,6)
l1=l1+1
```

Combination III

```
#Sensex with a sector
m1=matrix(c(1,1,1,1,1,1,1,0,1,1,0,1,1,1,0,0,1,0,1,1,1,0,1,0,1,0,0,1,1,0,0,0,0,1,1,1,0,1,1
,0,0,1,0,1,0,1,0,0,0,0,1,1,0,0,1,0,0,0,0,1,0,0,0,0),ncol=4,byrow=TRUE)
a1=expand.grid(1:0,1:0,1:0)
TP=matrix(rep(0,16*8),ncol=16)
for (i in 2:2474){
for (j in 1:8){
for (k in 1:16){
if(isTRUE(all.equal(se[i:(i+2),1],as.numeric(a1[j,])))) {
if(isTRUE(all.equal(as.numeric(se[i-1,14:17]),m1[k,]))) { TP[j,k]=TP[j,k]+1 }
} } } }
KP=matrix(rep(0,16*8),nrow=8)
for (i in 1:8) {
for (j in 1:16) {
KP[i,j]=TP[i,j]/sum(TP[i,])
} }
KP
```

Sensex past 3 days, Icici past 3 days

```
a1=expand.grid(1:0,1:0,1:0)
TP=matrix(rep(0,8*8),ncol=8)
SP=matrix(rep(0,8*8),ncol=8)
for (i in 1:2474){
for (j in 1:8){
for (k in 1:8){
if(isTRUE(all.equal(se[i:(i+2),1],as.numeric(a1[j,])))) {
if(isTRUE(all.equal(se[i:(i+2),6],as.numeric(a1[k,])))) {
TP[j,k]=TP[j,k]+1
if(se[i+3,6]==1) {  SP[j,k]=SP[j,k]+1  }
} } } } }
KP=matrix(rep(0,8*8),nrow=8)
for (i in 1:8) {
for (j in 1:8) {
KP[i,j]=SP[i,j]/TP[i,j]
} }
KP
```

```
#1d TPM 11,10,01,00
pr2=function(f2) {
n=(length(f2)-3)
zi=matrix(rep(0,4),ncol=4)
c1=0; p1=0; c2=0; p2=0
for (j in 1:n) {
if (is.na(f2[j])|is.na(f2[j+3])){
```

```
r=1
}
else{
if (f2[j]==1&f2[j+3]==1) {
c1=c1+1
}
if(f2[j+3]==1) {
p1=p1+1
}
if(f2[j]==0&f2[j+3]==1) {
c2=c2+1
}
if(f2[j+3]==0) {
p2=p2+1
} }
zi[1]=c1/p1; zi[2]=1-(c1/p1); zi[3]=c2/p2; zi[4]=1-(c2/p2)
}
return (zi) }

conmat2=function(ba,j=1,k=2) {
prop=matrix(c(0,0,0,0),nrow=2)
for (i in 1:(length(ba[,j])-3)) {
if(ba[i+3,j]==0&ba[i,k]==0) {
prop[2,2]=prop[2,2]+1
}
else if(ba[i+3,j]==1&ba[i,k]==1) {
prop[1,1]=prop[1,1]+1
}
else if(ba[i+3,j]==1&ba[i,k]==0) {
prop[2,1]=prop[2,1]+1
}
else {
prop[1,2]=prop[1,2]+1
} }
return (t(prop))
}

#HMM Combination 3
e1=conmat2(se,1,7)
e2=matrix(c(e1[1,1]/sum(e1[1,]),e1[1,2]/sum(e1[1,]),e1[2,1]/sum(e1[2,]),e1[2,2]/sum(
e1[2,])),nrow=2,byrow=TRUE)
hmm = initHMM(c("A","B"), c("L","R"),
transProbs=matrix(pr2(se[,7]),nrow=2,byrow=TRUE),
emissionProbs=e2)
print(hmm)
#Sequence of observations
```

```
observations = c("L","L","R","R")
#Calculate Viterbi path
viterbi = viterbi(hmm,observations)
print(viterbi)
```

A7. Code for Gamma coefficient

```
#across sectors TPM #a1 column indices eg. 5:8 for bank
ASTPM2=function(a1,b1) {
db=d[,a1]; da=d[,b1]
KP=matrix(rep(0,25),nrow=5); i=1
while(i<2477) {
k=ifelse(sum(da[i,])>=4,5,ifelse(sum(da[i,])>=3,4,ifelse(sum(da[i,])>=2,3,ifelse(sum(
da[i,])>=1,2,1))))
l=ifelse(sum(db[i,])>=4,5,ifelse(sum(db[i,])>=3,4,ifelse(sum(db[i,])>=2,3,ifelse(sum(
db[i,])>=1,2,1))))
i=i+1
KP[l,k]=KP[l,k]+1
}
return(KP)
}
Z5=ASTPM2(1:4,5:8)
colnames(Z5)=c('A','B','C','D','E')
rownames(Z5)=c('A','B','C','D','E')
write.csv(as.data.frame(Z5),"Contingency Table.csv")

#Gamma
Glam=matrix(rep(NA,64), nrow=8)
for (i in 1:8) {
for (j in 1:8) {
if(i<j) {
Glam[i,j]=GoodmanKruskalGamma(ASTPM2((4*i-3):(4*i),(4*j-3):(4*j)))
Glam[j,i]=Glam[i,j]
}
if(i==j) {
Glam[i,j]=1
} } }
Z6=round(Glam,3) #Gamma values across all sectors
write.csv(as.data.frame(Z6),"GKgamma.csv")

GoodmanKruskalGamma(ASTPM2(13:16,5:8))
```